• Easy • #136 • Single Number

▲ Problem Statement : Given an array where every element appears "twice" except for one element, find the element that appears only once.

• The solution must have $O(n)$ time & $O(1)$ extra space.

▲ Main Problem Solution : Use the XOR (^) operator.

• XOR has two important properties :- • $a \wedge a = 0$
• $a \wedge 0 = a$

• So when we XOR every element together, all duplicate elements cancel out & only the single number remains.

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Brute Force Method : Use a "hash map" to count the frequency of every element, then find the element with frequency "1".

Time → O(n)
Space → O(n)

Algorithm : 1|. Initialize "result = 0".

2|. Traverse through every element.

3|. XOR the current element with "result".

4|. Duplicate elements cancel each other out.

5|. The remaining value is the single number.

6|. Return "result"

Important Points :
- $x \wedge x = 0$
- $x \wedge 0 = x$

- XOR is commutative & associative, so order doesn't matter.

- All duplicate numbers cancel each other.

- Only the unique number remains.

Complexity : Time → O(n)
Space → O(1)